# INTERNET-OF-THINGS AND EMBEDDED SYSTEM PRACTICE

## Module Code: CSC83309

**LECTURE**

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# Chapter 6: IoT Connectivity and Network Protocol Stack

# 6.1 Overview of the IoT Connectivity Landscape

**Definition:**

IoT connectivity refers to the methods and technologies used to enable communication between IoT devices, gateways, and cloud or edge computing platforms.

**Key Requirements of IoT Connectivity:**

- Scalability: Support for millions of devices
- Low Power Consumption: Essential for battery-operated devices
- Wide Coverage: Ability to connect devices over large distances
- Low Latency: Important for real-time applications
- Reliability and Security: Ensuring data integrity and protection from attacks

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# 6.2 Communication Technologies for IoT

**Short-Range Wireless Technologies**

**Wi-Fi:**

- Widely used for home and office IoT devices.
- High data rates and good range (~100 meters).
- Higher power consumption compared to other protocols.
- Suitable for applications requiring high bandwidth.

**Bluetooth and Bluetooth Low Energy (BLE):**

- Short-range communication (~10 meters).
- BLE optimized for very low power consumption.
- Commonly used for wearable devices, health monitors, and smart home gadgets.

**Zigbee:**

- Mesh network topology allowing devices to relay data.
- Low power and low data rate (~250 kbps).
- Range up to 100 meters per node.
- Ideal for home automation and industrial sensor networks.

# Long-Range Wireless Technologies

**LoRa (Long Range):**

- Low-power wide-area network (LPWAN) technology.
- Range up to 10 km in rural areas.
- Low data rate (~0.3 to 50 kbps).
- Used for smart agriculture, smart cities, and asset tracking.

**NB-IoT (Narrowband IoT):**

- Cellular-based LPWAN technology.
- Operates in licensed spectrum bands.
- Suitable for massive IoT deployments with low throughput needs.
- Provides good indoor penetration and low power consumption.

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# Sigfox:

- Ultra-narrowband LPWAN technology.
- Very low data rates and power consumption.
- Ideal for simple sensor data transmission.

**Other Technologies**

- Ethernet: Wired connection used in industrial IoT for high reliability and speed.
- 5G: Emerging cellular technology offering ultra-low latency, high data rates, and massive device connectivity for IoT.

# 6.3 IoT Network Protocol Stack and Standards

| Layer | Function | Example Protocols |
|---|---|---|
| **Perception Layer** | Device identification and data sensing | IEEE 802.15.4 (physical/MAC layer) |
| **Network Layer** | Routing and data forwarding | IPv6, 6LoWPAN, RPL |
| **Transport Layer** | Reliable data transfer | TCP, UDP |
| **Application Layer** | Application-specific messaging | MQTT, CoAP, HTTP, DDS |

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**Connectivity Categories:**

- Short-Range Wireless: Suitable for personal area networks and home automation (e.g., Bluetooth, Zigbee).
- Wide-Area Wireless: Suitable for city-wide or regional connectivity (e.g., LoRaWAN, NB-IoT).
- Wired: Ethernet and other physical connections used in industrial settings.